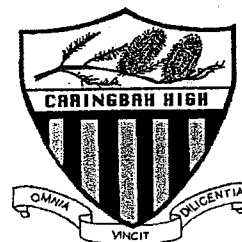


# **CARINGBAH HIGH SCHOOL**

**2010**

## **TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION**



## **Mathematics Extension 1**

### **General Instructions**

Reading time - 5 minutes

Working time - 2 hours

Write using black or blue pen.

Board-approved calculators may be used.

A table of standard integrals is provided at the back of this paper.

**Total marks - 84**

Attempt Questions 1 - 7

All questions of equal value.

All necessary working should be shown in every question.

Start each question on a new page.

**Question 1 (12 marks)****Marks**

(a) Factorise  $2x^3 + 54$ .

**2**

(b) Let  $f(x) = e^x - 1$ . What is the range of  $f(x)$ ?

**1**

(c) Given that  $\log_a b = 3 \cdot 4$  and  $\log_a c = 4 \cdot 5$ , find  $\log_a \left( \frac{b}{c} \right)$ .

**1**

(d) Differentiate  $e^{3x} \sin x$

**2**

(e) Find the exact value of  $\int_0^{\frac{\pi}{4}} \cos^2 x \, dx$ .

**3**

(f) Using the substitution  $u = x^2 + 1$ , or otherwise, find the

**3**

exact value of  $\int_0^3 \frac{2x}{x^2 + 1} \, dx$ .

**Question 2** (12 marks) Start a new page.

- (a) The function  $f(x) = x^3 - \log_e(x+1)$  has a root near  $x = 1$ . **3**

Use one application of Newton's method to obtain another approximation to this root. Give your answer correct to two decimal places.

- (b) The polynomial  $q(x) = x^3 + 3x^2 + ax + b$  has a factor of  $(x-2)$  and a remainder of  $-9$  when divided by  $(x+1)$ . **3**

Find the values of  $a$  and  $b$ .

- (c) (i) Express  $\sqrt{3} \sin x - \cos x$  in the form  $A \sin(x - \theta)$  where **2**  
 $0 \leq \theta \leq \frac{\pi}{2}$ .

- (ii) Hence, or otherwise, solve  $\sqrt{3} \sin x - \cos x = 1$  **2**  
for  $0 \leq x \leq 2\pi$ . Give your answer in exact form.

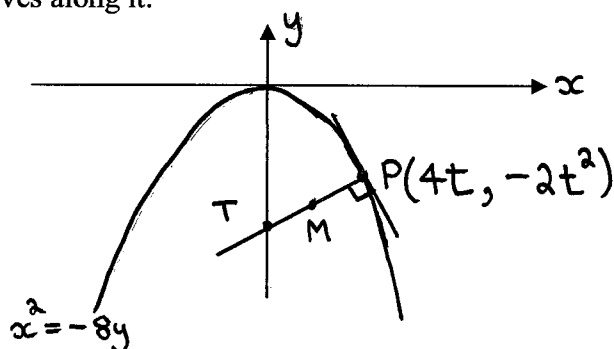
- (d) Evaluate  $\int_0^{\ln 3} e^{2x} dx$ . **2**

**Question 3** (12 marks) Start a new page.

(a) Find  $\lim_{x \rightarrow 0} \frac{\sin 5x}{2x}$

1

- (b) The diagram shows the parabola  $x^2 = -8y$  and the point  $P(4t, -2t^2)$  which moves along it.



- (i) Show that the equation of the normal at  $P$  is  $y = \frac{x}{t} - 2t^2 - 4$ . 2
- (ii) This normal meets the  $y$ -axis at  $T$ . The midpoint of  $PT$  is  $M$ . Find the coordinates of  $T$  and  $M$ . 2
- (iii) Find the Cartesian equation of the locus of  $M$ . 1
- (c) (i) On the same set of axes sketch the graphs of  $y = \cos 2x$  and  $y = -\frac{1}{2}x$  for  $0 \leq x \leq \pi$ . (Show at least 2 points which lie on  $y = -\frac{1}{2}x$ .) 2
- (ii) Use the graph to determine the number of solutions there are to the equation  $\cos 2x = -\frac{1}{2}x$  for  $0 \leq x \leq \pi$ . 1
- (d) Use mathematical induction to prove that  $7^n - 1$  is divisible by 3, for all integers  $n \geq 1$ . 3

**Question 4 (12 marks)** Start a new page.

- (a) A particle is moving in a straight line with its acceleration as a function of  $x$  given by  $\ddot{x} = 3x^2$ . It is initially 1 metre to the right of the origin and is travelling with a velocity of  $-\sqrt{2}$  metres per second.

(i) Show that  $\dot{x} = -\sqrt{2x^3}$  2

(ii) Hence show that  $x = \frac{2}{(t+\sqrt{2})^2}$  3

- (b) (i) Find the  $x$ -coordinates of stationary points on the curve 3

$$y = x - 2 \sin x \quad \text{for } 0 \leq x \leq 2\pi.$$

- (ii) Write down the  $y$ -coordinates of these stationary points in exact form. 1

- (iii) Find the nature of each of the stationary points. 1

- (iv) Find the coordinates of those points on the curve corresponding to  $x = 0$  and  $2\pi$ . 1

- (v) Hence sketch the curve for  $0 \leq x \leq 2\pi$ . (You do not have to find any other intercepts with the axes or points of inflexion.) 1

**Question 5** (12 marks) Start a new page.

- (a) The displacement  $x$  metres of a particle moving in simple harmonic motion is given by  $x = 5 \sin(\pi t + \frac{\pi}{2})$  where the time  $t$  is in seconds.
- (i) What is the period of the oscillation? 1
  - (ii) Show that the particle's oscillation starts 5 metres to the right of the equilibrium position. 1
  - (iii) What is the speed of the particle as it moves through the equilibrium position? 2
  - (iv) Show that the acceleration of the particle is proportional to the displacement from the equilibrium position. 2
- (b) Show that for  $0 < x < 1$ ,  $\frac{d}{dx} \left[ \sin^{-1} \sqrt{1-x^2} \right] = \frac{-1}{\sqrt{1-x^2}}$  3
- (c) (i) Sketch the graph of  $y = \tan x$  for  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ . 1
- (ii) By using (i), or otherwise, find those values of  $x$  2  
satisfying  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$  for which the geometric series  
 $1 + \sqrt{3} \tan x + 3 \tan^2 x + 3\sqrt{3} \tan^3 x + \dots$  has a limiting sum.

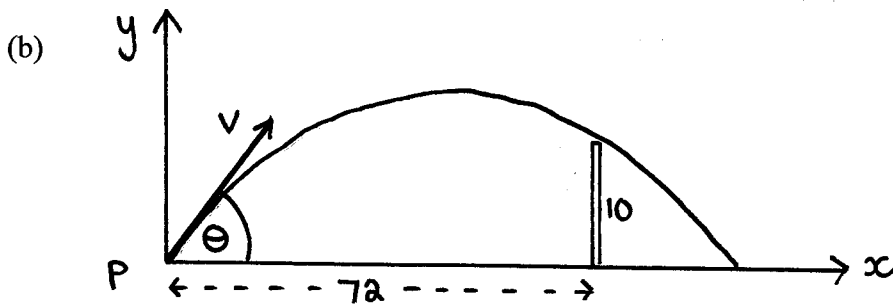
**Question 6 (12 marks)** Start a new page.

(a) Consider the function  $f(x) = x^2 - 4x + 7$ .

(i) Write down the largest positive domain for which  $f(x)$  has an inverse function  $f^{-1}(x)$ . 1

(ii) Find the inverse function  $f^{-1}(x)$ . 2

(iii) Sketch  $y = f^{-1}(x)$  and  $y = f(x)$  on the same diagram. 1



A ball is hit with initial velocity  $v$  from a point  $P$  on the ground at an angle of  $\theta$  with the horizontal.

(i) The equations of the motion of the ball are:

$$\ddot{x} = 0 \quad \text{and} \quad \ddot{y} = -10 \quad (\text{i.e. acceleration due to gravity is } 10\text{m/s}^2)$$

Using calculus, show that the position of the ball at time  $t$  is given by: 3

$$x = vt \cos \theta \quad \text{and} \quad y = vt \sin \theta - 5t^2$$

(ii) After 2 seconds the ball just clears a wall which is 10 metres high and 72 metres from  $P$ . 3

Calculate the angle of projection  $\theta$  (to the nearest degree) and the initial velocity  $v$ .

(iii) How far from  $P$  does the ball land? 2

**Question 7** (12 marks) Start a new page.

(a) Find  $\int \frac{1}{\sqrt{49 - x^2}} dx$ .

**1**

- (b) Nicole is staying in the Blue Mountains. She takes a cup of coffee whose temperature is  $80^\circ C$  outside where the air temperature is  $5^\circ C$ .

The temperature of the coffee,  $T^\circ C$ , after  $t$  minutes outside satisfies the equation

$$\frac{dT}{dt} = -k(T - 5).$$

- (i) Show that  $T = 5 + 75e^{-kt}$  satisfies both the above equation and the initial condition.

**2**

- (ii) After 10 minutes outside the temperature of the coffee is  $57^\circ C$ .  
After how many minutes (to the nearest minute) will the coffee's temperature be  $20^\circ C$ ?

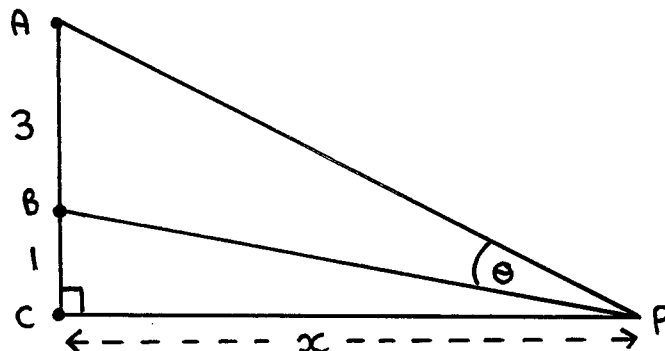
**3**

**Question 7 continues on page 9.**



**Question 7 (continued)**

(c)



In the diagram, the goal posts  $A$  and  $B$  in a children's soccer game are 3 metres apart. Goal post  $B$  is 1 metre from the corner post  $C$ . A player stands at  $P$  on the boundary line  $x$  metres from  $C$ . The goal posts  $A$  and  $B$  subtend an angle of  $\theta$  at  $P$ .

- (i) Show that  $\theta = \tan^{-1} \frac{4}{x} - \tan^{-1} \frac{1}{x}$ . 2
- (ii) Show that  $\theta$  is a maximum when  $x = 2$ . 2
- (iii) Deduce that the maximum angle subtended at  $P$  is  $\theta = \tan^{-1} \frac{3}{4}$ . 2

**End of paper**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, \quad x > 0$

# CHS EXT. 1 TRIAL HSC 2010

1. a)  $2(x^3 + 27) = 2(x+3)(x^2 - 3x + 9)$

b)  $y > -1$

c)  $\log_a\left(\frac{b}{c}\right) = \log_a b - \log_a c$   
 $= 3 \cdot 4 - 4 \cdot 5$   
 $= -1 \cdot 1$

d)  $u = e^{3x} \quad v = \sin x$   
 $u' = 3e^{3x} \quad v' = \cos x$

PRODUCT RULE:  $y' = vu' + uv'$   
 $= \sin x \cdot 3e^{3x} + e^{3x} \cdot \cos x$   
 $= e^{3x}(3\sin x + \cos x)$

e)  $\cos 2x = 2\cos^2 x - 1$

$\therefore \cos^2 x = \frac{1}{2}(1 + \cos 2x)$

$\int_0^{\frac{\pi}{4}} \cos^2 x = \frac{1}{2} \int_0^{\frac{\pi}{4}} 1 + \cos 2x \, dx$

$= \frac{1}{2} \left[ x + \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{4}}$

$= \frac{1}{2} \left[ \left( \frac{\pi}{4} + \frac{1}{2} \sin \frac{\pi}{2} \right) - \left( 0 + \frac{1}{2} \sin 0 \right) \right]$

$= \frac{\pi}{8} + \frac{1}{4}$

f)  $x=0: u=1$

$x=3: u=10$

$\frac{du}{dx} = 2x$

$\therefore du = 2x \, dx$

$\therefore \int_0^3 \frac{2x \, dx}{x^2 + 1} = \int_1^{10} \frac{du}{u}$   
 $= [\ln u]_1^{10}$   
 $= \ln 10 - \ln 1$   
 $= \ln 10$

2. a)  $f'(x) = 3x^2 - \frac{1}{x+1}$

$f'(1) = 3 - \frac{1}{2} = 2.5$

and  $f(1) = 1 - \ln 2 \approx 0.307$

$z_2 = z_1 - \frac{f(z_1)}{f'(z_1)}$  NEWTON'S METHOD

$= 1 - \frac{0.307}{2.5}$

$\approx 0.88$

b)  $(x-2)$  factor  $\therefore q(2)=0$

i.e.  $2a+b=-20$  ①

rem. = 9 when  $\div$  by  $(x+1)$   $\therefore q(-1)=-9$

i.e.  $-a+b=-11$  ②

Solving ① and ②:  $a=-3, b=-14$

c) i)  $\sqrt{(\sqrt{3})^2 + 1^2} = 2$

$\therefore \sqrt{3} \sin x - \cos x = 2 \left( \frac{\sqrt{3}}{2} \sin x - \frac{1}{2} \cos x \right)$

$= 2(\cos \theta \sin x - \sin \theta \cos x)$

$= 2 \sin(x-\theta)$  where  $\cos \theta = \frac{\sqrt{3}}{2}$   
 and  $\sin \theta = \frac{1}{2}$   
 i.e.  $\theta = \frac{\pi}{6}$

$= 2 \sin\left(x - \frac{\pi}{6}\right)$

ii)  $\sqrt{3} \sin x - \cos x = 1$

$2 \sin\left(x - \frac{\pi}{6}\right) = 1$

$\sin\left(x - \frac{\pi}{6}\right) = \frac{1}{2}$

$\left(x - \frac{\pi}{6}\right) = \frac{\pi}{6}, \frac{5\pi}{6}$

$x = \frac{\pi}{3}, \pi$

d)  $\frac{1}{2} [e^{2x}]_0^{\ln 3} = \frac{1}{2} [e^{2 \ln 3} - e^0]$

$= \frac{1}{2} [e^{\ln 9} - 1]$

$= \frac{1}{2} [9 - 1]$

$= 4$

3. a)  $\lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \times \frac{5}{2} = 1 \times \frac{5}{2}$   
 $= 2\frac{1}{2}$

$$3 \text{ (b) } y = -\frac{1}{8}x^2$$

$$y' = -\frac{1}{4}x$$

$$\text{At } (4t, -2t^2): m_{\text{TAN}} = -\frac{1}{4} \times 4t = -t$$

$$\therefore m_{\text{NORM}} = \frac{1}{t}$$

$$\text{EQUON NORM: } y + 2t^2 = \frac{1}{t}(x - 4t)$$

$$y = \frac{x}{t} - 2t^2 - 4$$

$$\text{(ii) } T(0, -2t^2 - 4)$$

$$M: x = \frac{4t+0}{2}, y = \frac{-2t^2 + -2t^2 - 4}{2}$$

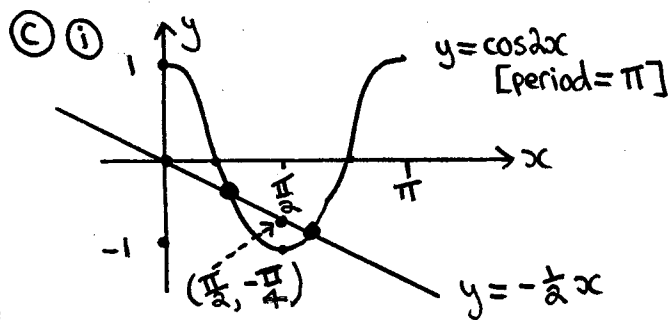
$$= 2t, \quad = -2t^2 - 2$$

$$\text{i.e. } M(2t, -2t^2 - 2)$$

$$\text{(ii) sub } t = \frac{x}{2} \text{ into } y = -2t^2 - 2$$

$$y = -2 \times \frac{x^2}{4} - 2$$

$$= -\frac{1}{2}x^2 - 2$$



$$\text{(ii) 2 points of intersection.}$$

$$\therefore \cos 2x = -\frac{1}{2}x \text{ has 2 solns}$$

$$\text{(d) STEP 1: Prove true for } n=1$$

$$7^1 - 1 = 6 \text{ which is divis. by 3}$$

$$\text{STEP 2: Assume true for } n=k$$

$$\text{i.e. } 7^k - 1 = 3M \text{ (M an integer)}$$

$$[\text{OR } 7^k = 3M + 1]$$

$$\text{Hence prove true for } n=k+1$$

$$\text{i.e. } 7^{k+1} - 1 \text{ also divis. by 3}$$

$$\text{Now } 7^{k+1} - 1 = 7 \cdot 7^k - 1$$

$$= 7 \cdot (3M + 1) - 1 \text{ by our assumption}$$

$$= 21M + 7 - 1$$

$$= 21M + 6$$

$$= 3(7M + 2) \text{ which is divis. by 3}$$

$$\text{i.e. if true for } n=k \text{ then true for } n=k+1$$

$$\text{STEP 3: we assumed true for } k \text{ and hence proved true for } n=k+1.$$

$$\text{Since true for } n=1 \text{ then by the Principle of Mathematical Induction true for all positive integers.}$$

$$4. \text{ (a) (i) } \frac{d}{dx}(\frac{1}{2}v^2) = 3x^2$$

$$\frac{1}{2}v^2 = x^3 + c$$

$$\text{Sub } x=1, v=-\sqrt{2}: 1 = 1 + c \Rightarrow c = 0$$

$$\frac{1}{2}v^2 = x^3$$

$$v^2 = 2x^3$$

$$v = \pm \sqrt{2x^3} \text{ but take -ve since } v = -\sqrt{2} \text{ initially}$$

$$v = -\sqrt{2x^3}$$

$$\text{(ii) } \frac{dx}{dt} = -\sqrt{2x^3}$$

$$\frac{dt}{dx} = -\frac{1}{\sqrt{2x^3}} = -\frac{1}{\sqrt{2}} x^{-3/2}$$

$$t = -\frac{1}{\sqrt{2}} \cdot 2x^{-1/2} + c$$

$$= \frac{\sqrt{2}}{\sqrt{x}} + c$$

$$\text{Sub } t=0, x=1: 0 = \sqrt{2} + c$$

$$c = -\sqrt{2}$$

$$\text{i.e. } t = \frac{\sqrt{2}}{\sqrt{x}} - \sqrt{2}$$

$$t + \sqrt{2} = \frac{\sqrt{2}}{\sqrt{x}}$$

$$(t + \sqrt{2})^2 = \frac{2}{x}$$

$$x = \frac{2}{(t + \sqrt{2})^2}$$

$$\text{(b) (i) } \frac{dy}{dx} = 1 - 2\cos x$$

$$\text{Stationary points } \frac{dy}{dx} = 0:$$

$$1 - 2\cos x = 0$$

$$\cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3}, \frac{5\pi}{3}$$

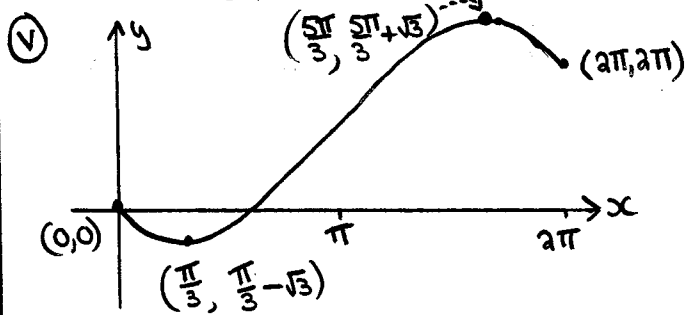
4. (b) (ii)  $x = \frac{\pi}{3}$ ,  $y = \frac{\pi}{3} - 2 \times \frac{\sqrt{3}}{2} = \frac{\pi}{3} - \sqrt{3}$   
 $x = \frac{5\pi}{3}$ ,  $y = \frac{5\pi}{3} - 2 \times -\frac{\sqrt{3}}{2} = \frac{5\pi}{3} + \sqrt{3}$

(iii)  $\frac{d^2y}{dx^2} = 2 \sin x$

$(\frac{\pi}{3}, \frac{\pi}{3} - \sqrt{3})$ :  $\frac{d^2y}{dx^2} = 2 \sin \frac{\pi}{3} = \sqrt{3} > 0$   
 $\therefore$  min. turning point

$(\frac{5\pi}{3}, \frac{5\pi}{3} + \sqrt{3})$ :  $\frac{d^2y}{dx^2} = 2 \sin \frac{5\pi}{3} = -\sqrt{3} < 0$   
 $\therefore$  max. turning point

(iv)  $x=0$ :  $y = 0 - 2 \sin 0 = 0$  i.e.  $(0,0)$   
 $x=2\pi$ :  $y = 2\pi - 2 \sin 2\pi = 2\pi$  i.e.  $(2\pi, 2\pi)$



5. (a) (i) period =  $\frac{2\pi}{n} = \frac{2\pi}{\pi} = 2s$

(ii) sub  $t=0$ :  $x = 5 \sin \frac{\pi}{2} = 5 \times 1 = 5$  i.e. 5m right of  $x=0$ .

(iii) when  $x=0$ :  $5 \sin(\pi t + \frac{\pi}{2}) = 0$

$\pi t + \frac{\pi}{2} = 0, \pi, \dots$

$\pi t = -\frac{\pi}{2}, \frac{\pi}{2}, \dots$

$t = -\frac{1}{2}, \frac{1}{2}, \dots$  ignore  $t = -\frac{1}{2}$

i.e. thru  $x=0$  when  $t = \frac{1}{2}s$

$v = \dot{x} = 5\pi \cos(\pi t + \frac{\pi}{2})$

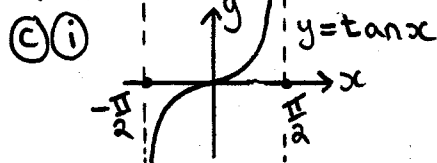
when  $t = \frac{1}{2}$ :  $v = 5\pi \cos(\frac{\pi}{2} + \frac{\pi}{2}) = 5\pi \cos \pi = -5\pi$  m/s

i.e. speed =  $5\pi$  m/s

(iv)  $a = \ddot{x} = -5\pi^2 \sin(\pi t + \frac{\pi}{2}) = -\pi^2 \times 5 \sin(\pi t + \frac{\pi}{2}) = -\pi^2 x$

i.e. acceleration proportional to displacement

(b)  $\frac{1}{\sqrt{1-(\sqrt{1-x^2})^2}} \times \frac{-x}{\sqrt{1-x^2}}$   
 $= \frac{1}{\sqrt{1-(1-x^2)}} \times \frac{-x}{\sqrt{1-x^2}}$   
 $= \frac{1}{\sqrt{x^2}} \times \frac{-x}{\sqrt{1-x^2}}$   
 $= \frac{-1}{\sqrt{1-x^2}}$   
 since  $\frac{d}{dx} (1-x^2)^{\frac{1}{2}} = \frac{1}{2} (1-x^2)^{-\frac{1}{2}} \cdot -2x = \frac{-x}{\sqrt{1-x^2}}$



(ii) series has limiting sum if  $-1 < r < 1$ .

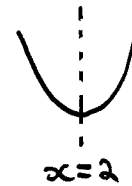
Here  $r = \sqrt{3} \tan x$

i.e.  $-1 < \sqrt{3} \tan x < 1$

$-\frac{1}{\sqrt{3}} < \tan x < \frac{1}{\sqrt{3}}$

$\therefore -\frac{\pi}{6} < x < \frac{\pi}{6}$

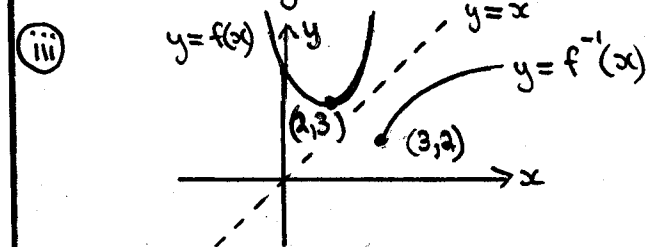
6. (a) (i) axis of symmetry:  $x = -\frac{-4}{2} = 2$



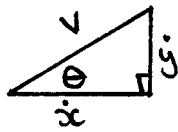
$\therefore$  DOMAIN  $x \geq 2$

(ii) swap  $x, y$ :  $x = y^2 - 4y + 7$   
 $x - 7 = y^2 - 4y$   
 $x - 7 + 4 = y^2 - 4y + 4$   
 $x - 3 = (y - 2)^2$   
 $y - 2 = \pm \sqrt{x - 3}$   
 $y = 2 \pm \sqrt{x - 3}$

TAKE  $y = 2 + \sqrt{x - 3}$  SINCE RANGE OF  $f^{-1}(x)$  IS  $y \geq 2$



6. (b) (i) INITIALLY



$$\dot{x} = v \cos \theta$$

$$\dot{y} = v \sin \theta$$

HORIZONTAL

$$\ddot{x} = 0$$

$$\dot{x} = c$$

$$\therefore \dot{x} = v \cos \theta$$

$$x = vt \cos \theta + k$$

$$\text{sub } t=0, x=0 \Rightarrow k=0$$

$$x = vt \cos \theta$$

VERTICAL

$$\ddot{y} = -10$$

$$\dot{y} = -10t + c$$

$$\text{sub } t=0, \dot{y} = v \sin \theta$$

$$\Rightarrow c = v \sin \theta$$

$$\dot{y} = v \sin \theta - 10t$$

$$y = vt \sin \theta - 5t^2 + k$$

$$\text{sub } t=0, y=0 \Rightarrow k=0$$

$$y = vt \sin \theta - 5t^2$$

(ii) sub  $t=2, x=72$  into  $x = vt \cos \theta$

$$72 = 2v \cos \theta \quad (1)$$

sub  $t=2, y=10$  into  $y = vt \sin \theta - 5t^2$

$$10 = 2v \sin \theta - 20$$

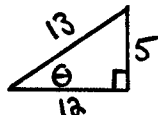
$$30 = 2v \sin \theta \quad (2)$$

$$(2) \div (1): \frac{2v \sin \theta}{2v \cos \theta} = \frac{30}{72}$$

$$\tan \theta = \frac{5}{12}$$

$$\theta \doteq 23^\circ$$

since  $\tan \theta = \frac{5}{12} \therefore \cos \theta = \frac{12}{13}$



$$[\sin \theta = \frac{5}{13}]$$

so (1):  $72 = 2v \times \frac{12}{13}$

$$v = 39 \text{ m/s}$$

(ii)  $y = 39t \times \frac{5}{13} - 5t^2 = 15t - 5t^2$   
hits ground when  $y=0: 15t - 5t^2 = 0$   
 $5t(3-t) = 0$   
 $t = 0, 3.$

$$\therefore \text{dist. from P: } x = 39 \times 3 \times \frac{12}{13} = 108 \text{ m}$$

7. (a)  $\sin^{-1} \frac{x}{7} + c$

(b) (i)  $T = 5 + 75e^{-kt}$

$$\frac{dT}{dt} = -k \times 75e^{-kt}$$

$$= -k(T-5) \text{ since } T-5 = 75e^{-kt}$$

also when  $t=0: T = 5 + 75e^0 = 80$

(ii) sub  $t=10, T=57: 57 = 5 + 75e^{-10k}$

$$52 = 75e^{-10k}$$

$$\frac{52}{75} = e^{-10k}$$

$$k = \ln\left(\frac{52}{75}\right) \div -10$$

$$\doteq 0.0366$$

sub  $T=20: 20 = 5 + 75e^{-0.0366t}$

$$15 = 75e^{-0.0366t}$$

$$0.2 = e^{-0.0366t}$$

$$t = \ln 0.2 \div -0.0366$$

$$\doteq 44 \text{ min}$$

(c) (i) Let  $\angle BPC = \alpha$

In  $\triangle ACP: \tan(\theta + \alpha) = \frac{4}{x}$

$$\therefore \theta + \alpha = \tan^{-1} \frac{4}{x}$$

In  $\triangle BCP: \tan \alpha = \frac{1}{x}$

$$\therefore \alpha = \tan^{-1} \frac{1}{x}$$

Now  $\theta = (\theta + \alpha) - \alpha = \tan^{-1} \frac{4}{x} - \tan^{-1} \frac{1}{x}$

(ii)  $\frac{d\theta}{dx} = \frac{1}{1 + \left(\frac{4}{x}\right)^2} \times -\frac{4}{x^2} - \frac{1}{1 + \left(\frac{1}{x}\right)^2} \times -\frac{1}{x^2}$   
$$= \frac{-4}{x^2 + 16} + \frac{1}{x^2 + 1}$$

when  $x=2: \frac{d\theta}{dx} = \frac{-4}{20} + \frac{1}{5} = 0$

$\therefore$  stationary point when  $x=2.$

$$\frac{d^2\theta}{dx^2} = \frac{8x}{(x^2 + 16)^2} - \frac{2x}{(x^2 + 1)^2}$$

when  $x=2: \frac{d^2\theta}{dx^2} = \frac{16}{400} - \frac{4}{25} = -\frac{3}{25} < 0$

$\therefore \theta$  max. when  $x=2.$

(iii) So max.  $\theta = \tan^{-1} 2 - \tan^{-1} \frac{1}{2}$

let  $x = \tan^{-1} 2 \therefore \tan x = 2$

and  $y = \tan^{-1} \frac{1}{2} \therefore \tan y = \frac{1}{2}$

i.e.  $\theta = x - y$

$$\tan \theta = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$= \frac{2 - \frac{1}{2}}{1 + 2 \times \frac{1}{2}}$$

$$= \frac{3}{4}$$

$$\therefore \theta = \tan^{-1} \frac{3}{4}$$